

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-I]

TEST # 17

DATE : 05-01-2020

REVISION TEST

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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REVISION TEST # 17

Topics

DATE : 05 - 01 - 2020**PRE-MEDICAL ENTHUSIAST COURSE : PHASE - I – PATTERN : [NEET]****PHYSICS** : Modern Physics**CHEMISTRY** : d & bf-Block, Environmental Chemistry**ZOOLOGY** : Cockroach, Earthworm, Frog, Human Reproduction (Gametogenesis to Parturition)

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

- | | |
|---|--|
| <p>1. In Young's experiment, the separation between 5th maxima and 3rd minima is how many times as that of fringe width?
 (1) 5 times (2) 3 times
 (3) 2.5 times (4) 2 times</p> <p>2. It is found that when waves from two identical coherent sources superpose at a certain point, the resultant intensity is equal to intensity of one wave only. This means that the phase difference between the two waves at that point is
 (1) Zero (2) $\pi/3$
 (3) $2\pi/3$ (4) π</p> <p>3. In Young's double slit experiment, when two light waves form third minimum, they have
 (1) Phase difference of 3π
 (2) Path difference of 3λ

 (3) Phase difference of $\frac{5\pi}{2}$

 (4) Path difference of $\frac{5\lambda}{2}$</p> <p>4. In a Young's double slit experiment, the fringe width is found to be 0.4 mm. If the whole apparatus is immersed in water of refractive index $\frac{4}{3}$ without disturbing the geometrical arrangement, the new fringe width will be
 (1) 0.30 mm (2) 0.40 mm
 (3) 0.53 mm (4) 450 μm</p> | <p>5. Consider an YDSE that has different slits width, as a result, amplitudes of waves from two slits are A and 2A, respectively. If I_0 be the maximum intensity of the interference pattern, then intensity of the pattern at a point where phase difference between waves is ϕ, is :-

 (1) $I_0 \cos^2 \phi$ (2) $\frac{I_0}{3} \sin^2 \frac{\phi}{2}$

 (3) $\frac{I_0}{9} [5 + 4 \cos \phi]$ (4) $\frac{I_0}{9} [5 + 8 \cos \phi]$</p> <p>6. In a standard YDSE setup, two coherent sources of light of intensity ratio β produces interference pattern. $\frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}$ will be equal to ($I_{\max}$ & $I_{\min}$ are maximum and minimum intensities of the resultant wave)

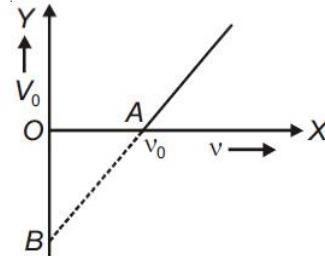
 (1) $\frac{2\sqrt{\beta}}{1-\beta}$ (2) $\frac{2\sqrt{\beta}}{1+\beta}$

 (3) $\frac{2\beta}{1+\beta}$ (4) $\frac{\sqrt{\beta}}{1+\beta}$</p> <p>7. The photoelectrons emitted from a surface of sodium metal are such that they
 (1) All are of the same frequency
 (2) Have the same kinetic energy
 (3) Have the same de Broglie wavelength
 (4) Have their speeds varying from zero to a certain maximum</p> |
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8. A photo cell is receiving light from a source placed at a distance of 1 m. If the same source is to be placed at a distance of 2 m, then the ejected electron
- Moves with one-fourth energy as that of the initial energy
 - Moves with one-fourth of momentum as that of the initial momentum
 - Will be half in number
 - Will be one-fourth in number
9. Stopping potential for photo electrons
- Does not depend on the frequency of the incident light
 - Does not depend upon the nature of the cathode material
 - Depends on both the frequency of the incident light and nature of the cathode material
 - Depends upon the intensity of the incident light
10. Assuming photo-emission to take place, the factor by which the maximum velocity of the emitted photo electrons changes when the wavelength of the incident radiation is increased four times, is (assuming work function to be negligible in comparison to hc/λ)
- 4
 - $\frac{1}{4}$
 - 2
 - $\frac{1}{2}$
11. The maximum wavelength of radiation that can produce photoelectric effect in a certain metal is 200 nm. The maximum kinetic energy acquired by electron due to radiation of wavelength 100 nm will be
- 12.4 eV
 - 6.2 eV
 - 100 eV
 - 200 eV

12. In an experiment on photoelectric effect the frequency f of the incident light is plotted against the stopping potential V_0 . The work function of the photoelectric surface is given by (e is electronic charge)

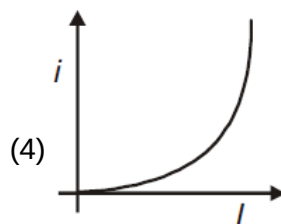
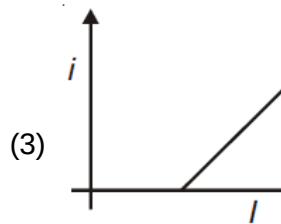
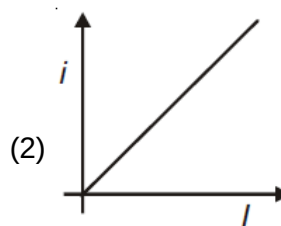
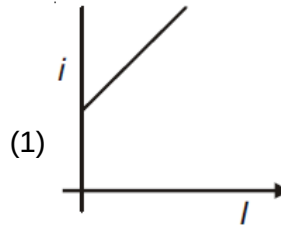


- $OB \times e$ in eV
 - OB in volt
 - OA in eV
 - The slope of the line AB
13. The threshold frequency for a metallic surface corresponds to an energy of 6.2 eV and the stopping potential for a radiation incident on this surface is 5 V. The incident radiation lies in
- Ultraviolet region
 - Infrared region
 - Visible region
 - X-ray region
14. A red bulb and violet bulb of equal power emits n_R and n_V number of photons in a given time, then
- $n_R = n_V$
 - $n_R > n_V$
 - $n_R < n_V$
 - $n_R \geq n_V$
15. For a certain metal, v is five times the v_0 and the maximum velocity of coming out photo electrons is 8×10^6 . If $v = 2v_0$, the maximum velocity of photo electrons will be
- 4×10^6 m/s
 - 6×10^6 m/s
 - 8×10^6 m/s
 - 1×10^6 m/s

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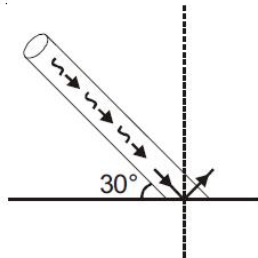
16. Two sources A and B have same power. The wavelength of radiation of A is λ_a and that of B is λ_b . The number of photons emitted per second by A and B are n_a and n_b respectively, then
- If $\lambda_a < \lambda_b$, $n_a = n_b$
 - If $\lambda_a > \lambda_b$, $n_a < n_b$
 - If $\lambda_a < \lambda_b$, $n_a < n_b$
 - If $\lambda_a > \lambda_b$, $n_a = n_b$
17. The work function of a certain metal is 2.3 eV. If light of wave number $2 \times 10^6 \text{ m}^{-1}$ falls on it, the kinetic energies of fastest and slowest ejected electron will be respectively
- 2.48 eV, 0.18 eV
 - 0.18 eV, zero
 - 2.30 eV, 0.18 eV
 - 0.18 eV, 0.18 eV
18. When the electromagnetic radiations of frequencies $4 \times 10^{15} \text{ Hz}$ and $6 \times 10^{15} \text{ Hz}$ fall on the same metal in different experiments, the ratio of maximum kinetic energy of electrons liberated is 1 : 3. The threshold frequency for the metal is
- $2 \times 10^{15} \text{ Hz}$
 - $1 \times 10^{15} \text{ Hz}$
 - $3 \times 10^{15} \text{ Hz}$
 - $1.67 \times 10^{15} \text{ Hz}$
19. The number of photons emitted by a 60 W bulb per second, if 10% of the electrical energy supplied to an incandescent light bulb is radiated as visible light, ($\lambda = 6000 \text{ \AA}$) is
- | | |
|--------------------------|--------------------------|
| (1) 1.8×10^{19} | (2) 1.8×10^{16} |
| (3) 1.8×10^{11} | (4) 1.8×10^{21} |

20. The graph between intensity of light falling on a metallic plate (I) with the current (i) generated is



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21. The number of photons ($\lambda = 6630 \text{ \AA}$) that strike per second on a totally reflecting screen (as shown in figure), so that a force of 1 N is exerted on the screen, is approximately

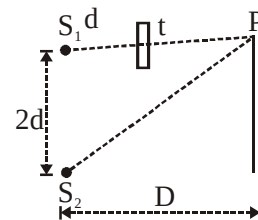


- (1) 10^{23} (2) 10^{27}
(3) 10^{25} (4) 10^{26}
22. In an interference pattern, at a point we observe the 16th order bright fringe or maxima for $\lambda_1 = 7200 \text{ \AA}$. What order will be visible here if the source is replaced by light of wavelength $\lambda_2 = 4800 \text{ \AA}$.
- (1) 12 (2) 24
(3) 36 (4) 16
23. In the Young's double slit experiment, the interference pattern is found to have an intensity ratio between bright and dark fringes, as 9. This implies that :-
- (1) the intensities at the screen due to the two slits are 5 units and 1 unit respectively
(2) the intensities at the screen due to the two slits are 4 units and 1 unit respectively
(3) the amplitude ratio of sources is 3
(4) the amplitude ratio of sources is 2

24. In Young's double slit interference experiment, the distance between two sources is 0.1 mm. The distance of the screen from the sources is 20 cms. Wavelength of light used is $5460 \text{ \AA}$. Then the angular position of the first dark fringe is :-

- (1) 0.08° (2) 0.16°
(3) 0.20° (4) 0.32°

25. If a thin mica sheet of thickness t and refractive index $\mu = (5/3)$ is placed in the path of one of the interfering beams as shown in figure, then the displacement of the fringe system is :-



- (1) $\frac{Dt}{3d}$ (2) $\frac{Dt}{5d}$
(3) $\frac{Dt}{4d}$ (4) $\frac{2Dt}{5d}$

26. A thin sheet of glass ($\mu = 1.5$) of thickness 6 microns introduced in the path of one of interfering beams in a double slit experiment shifts the central fringe to a position previously occupied by fifth bright fringe. Then the wavelength of light used is :-

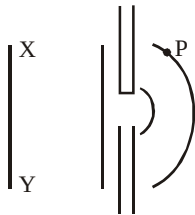
- (1) $6000 \text{ \AA}$ (2) $3000 \text{ \AA}$
(3) $4500 \text{ \AA}$ (4) $7500 \text{ \AA}$

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27. When light wave travelling in air suffers reflection at the interface between air and glass, the change of phase of the reflected wave is equal to :-

- (1) zero (2) $\pi/2$
(3) π (4) 2π

28. A monochromatic plane wave of speed c and wavelength λ is diffracted at a small aperture. The diagram illustrates successive wavefronts. After what time will some portion of the wavefront XY reach P ?



- (1) $\frac{3\lambda}{2c}$ (2) $\frac{2\lambda}{c}$ (3) $\frac{3\lambda}{c}$ (4) $\frac{4\lambda}{c}$

29. Some currency notes (to avoid counterfeits) change their colour as you tilt them. This is due to :-

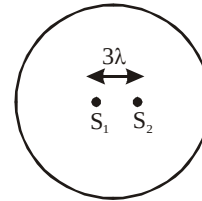
- (1) Diffraction (2) Polarization
(3) Interference (4) Refraction

30. In Young's double slit experiment, the intensity of light at a point on the screen where the path difference is λ is I_0 . The intensity of light at a point where the path

difference becomes $\frac{\lambda}{3}$ is:-

- (1) I_0 (2) $\frac{I_0}{4}$
(3) $\frac{I_0}{3}$ (4) $\frac{I_0}{2}$

31. If two coherent sources are placed at a distance 3λ from each other symmetric to the centre of the circle shown in the figure, then number of maxima shown on the screen placed along the circumference is :-



- (1) 16 (2) 12 (3) 8 (4) 4

32. Light waves can be polarised because they :-

- (1) have high frequencies
(2) have short wavelength
(3) are transverse
(4) can be reflected

33. A ray of light strikes a glass plate at an angle of 60° . If the reflected and refracted rays are perpendicular to each other, the index of refraction of glass is :-

- (1) $\sqrt{3}$ (2) $3/2$
(3) $\sqrt{3/2}$ (4) $1/2$

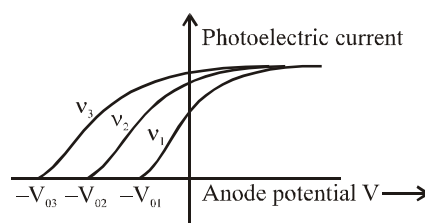
34. Choose the correct statement

- (1) the Brewster's angle is independent of wavelength of light
(2) the Brewster's angle is independent of nature of reflection surface
(3) the Brewster's angle is different for different wavelengths
(4) Brewster's angle depends on wavelength but not on the nature of reflecting surface

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35. Two point separates by distance of 0.1 mm can just be inspected in a microscope when light of wavelength $6000\text{\AA}$ is used, when light of wavelength $4800\text{\AA}$ is used then limit of resolution will become
- 0.80 mm
 - 0.12 mm
 - 0.10 mm
 - 0.08 mm
36. The aperture of the largest telescope in the world is 5 m, if the separation between the moon and the earth is 4×10^5 km and the wavelength of the visible light is $5000\text{\AA}$ then the minimum separation between objects on the surface of the moon which can be just resolve is approximately
- 1 m
 - 10 m
 - 50 m
 - 200 m
37. Consider a parallel beam of light of wavelength 600 nm and intensity 100 W/m^2 . How many photons cross 1cm^2 area perpendicular to beam in one second ?
- 3×10^{16}
 - 3×10^{14}
 - 3×10^9
 - 3×10^4

38. The photoelectric work function
- Is different for different materials
 - Is same for all metals
 - Depends upon frequency of the incident light
 - Depends upon intensity of the incident light
39. In photoelectric effect, the curve between photoelectric current and anode potential V (For different frequencies) is shown in figure, then



- $v_1 > v_2 > v_3$
 - $v_1 < v_2 < v_3$
 - $v_1 = v_2 = v_3$
 - $v_1 > v_2 < v_3$
40. The number of photo-electrons emitted per second from a metal surface increases when
- The energy of incident photons increases
 - The frequency of incident light increases
 - The wavelength of incident light increases
 - The intensity of incident light increases

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| <p>41. Wave nature of light cannot explain photoelectric effect because in photoelectric effect, it is seen that :</p> <p>(1) For the frequency of light below a certain value, the photoelectric effect does not take place, irrespective of intensity</p> <p>(2) Maximum kinetic energy of ejected electrons is independent of intensity of radiation</p> <p>(3) There is no time lag between the incidence of radiation and emission of electrons</p> <p>(4) All of these</p> <p>42. When photon of energy 3.8 eV falls on metallic surface of work function 2.8 eV, then the kinetic energy of emitted electrons are :-</p> <p>(1) 1 eV (2) 6.6 eV</p> <p>(3) 0 to 1 eV (4) 2.8 eV</p> <p>43. A monochromatic point source of light is placed at a distance d from a metal surface. Photoelectrons are ejected at a rate n per second, and with maximum kinetic energy E. If the source is brought nearer to distance $d/2$, the rate and the maximum kinetic energy per photoelectron become nearly</p> <p>(1) $2n$ and $2E$</p> <p>(2) $4n$ and $4E$</p> <p>(3) $4n$ and E</p> <p>(4) n and $4E$</p> | <p>44. If in a photoelectric experiment the wavelength of incident radiation is reduced from $6000 \text{ \AA}$ to $4000 \text{ \AA}$ then :</p> <p>(1) Stopping potential will decrease</p> <p>(2) Stopping potential will increase</p> <p>(3) Kinetic energy of emitted electrons will decrease</p> <p>(4) The value of work function will decrease</p> <p>45. Two identical photocathodes receive light of frequencies f_1 and f_2. If the velocities of the photo electrons (of mass m) coming out are respectively v_1 and v_2 then</p> <p>(1) $f_1 - f_2 = \frac{m(v_1^2 - v_2^2)}{2h}$</p> <p>(2) $f_1 + f_2 = \frac{m(v_1^2 - v_2^2)}{2h}$</p> <p>(3) $f_1 - f_2 = \frac{m(v_1^2 + v_2^2)}{2h}$</p> <p>(4) $f_1 + f_2 = \frac{m(v_1^2 + v_2^2)}{2h}$</p> |
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46. Which element is having lowest melting and boiling point :
(1) Ti (2) Cu (3) Zn (4) Mn
47. Density of which of the following element is highest :
(1) Pt (2) Hg
(3) Mn (4) Cu
48. The adsorption of hydrogen by platinum black is called :
(1) hydrogenation
(2) Reduction
(3) Occlusion
(4) Hydration
49. To which of the following series the transition element from $Z = 39$ to $Z = 48$ belong :
(1) 3d series (2) 4d series
(3) 5d series (4) 6d series
50. Chromium forms most stable compound in the following oxidation state :
(1) Cr (I) (2) Cr (II)
(3) Cr (III) (4) Cr (IV)
51. Which of the following is ferromagnetic :
(1) Cu, Ag, Au
(2) Fe, Co, Ni
(3) Zn, Cd, Hg
(4) Ca, Sr, Ba
52. $N_2(g) + 3H_2(g) \xrightleftharpoons{Fe + Mo} 2NH_3(g)$; Haber's process, Mo is used as :
(1) A catalyst
(2) A catalytic promoter
(3) An oxidising agent
(4) As a catalytic poison
53. $Cr_2O_7^{2-} \xrightleftharpoons[X]{Y} 2CrO_4^{2-}$, X and Y are respectively :
(1) $X = OH^-$, $Y = H^+$
(2) $X = H^+$, $Y = OH^-$
(3) $X = OH^-$, $Y = H_2O_2$
(4) $X = H_2O_2$, $Y = OH^-$
54. During estimation of oxalic acid Vs $KMnO_4$, self indicator is :
(1) $KMnO_4$ (2) Oxalic acid
(3) K_2SO_4 (4) $MnSO_4$
55. Bayer's reagent used to detect olifinic double bond is :
(1) Acidified $KMnO_4$
(2) Aqueous $KMnO_4$
(3) 1% alkaline $KMnO_4$ solution
(4) $KMnO_4$ in benzene
56. $Cu + conc. HNO_3 \rightarrow Cu(NO_3)_2 + X$ (oxide of nitrogen); (hot)
then X is :
(1) N_2O (2) NO_2 (3) NO (4) N_2O_3
57. The higher oxidation states of transition elements are found to be the combination with A and B, which are:
(1) F, O (2) O, N (3) O, Cl (4) F, Cl
58. Manganese steel is used for making railways because :
(1) It is hard with high percentage of Mn
(2) It is soft with high percentage of Mn
(3) It is hard with small concentration of manganese with the impurities
(4) It is soft with small concentration of manganese with impurities.

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59. 'Bordeaux mixture' is used as a fungicide. It is a mixture of :
 (1) $\text{CaSO}_4 + \text{Cu}(\text{OH})_2$
 (2) $\text{CuSO}_4 + \text{Ca}(\text{OH})_2$
 (3) $\text{CuSO}_4 + \text{CaO}$
 (4) $\text{CuO} + \text{CaO}$
60. Anhydrous ferric chloride is prepared by :
 (1) heating hydrated ferric chloride at a high temperature in a stream of air
 (2) heating metallic iron in a stream of dry chlorine gas
 (3) reaction of ferric oxide with HCl (aq)
 (4) reaction of metallic iron with HCl (aq)
61. Copper becomes green when exposed to moist air for longer period.
 (1) Because of the formation of a layer of cupric oxide on the surface of copper
 (2) Because of the formation of a layer of basic carbonate of copper on the surface of copper
 (3) Because of the formation of a layer of cupric hydroxide on the surface of copper.
 (4) (1) and (3) both
62. $\text{FeCl}_3 \cdot 6\text{H}_2\text{O} + \text{C}(\text{CH}_3)_2 (\text{CH}_3\text{O})_2 \rightarrow \text{Products}$
 Reaction products are
 (1) FeCl_3 , CH_3OH and CH_3COCH_3
 (2) $(\text{CH}_3\text{O})_3\text{Fe}$, HCl and H_2O
 (3) FeCl_2 , HCl and CH_3COCH_3
 (4) $\text{Fe}(\text{OH})_3$, FeCl_3 and CH_3COCH_3
63. When KI (excess) is added to :
 I : CuSO_4 II. HgCl_2 III. $\text{Pb}(\text{NO}_3)_2$
 (1) A white ppt. of CuI in I, an orange ppt. HgI_2 in II and a yellow ppt. of PbI_2 in III
 (2) A white ppt. of CuI in I, an orange ppt. dissolving to HgI_4^{2-} in II, and a yellow ppt. of PbI_2 in III
 (3) A white ppt. of CuI , HgI_2 and PbI_2 in each case
 (4) none is correct
64. CuSO_4 can be estimated volumetrically :
 (1) by reaction with KI followed by reaction with $\text{Na}_2\text{S}_2\text{O}_3$
 (2) by reaction with BaCl_2
 (3) by reaction with $\text{K}_4[\text{Fe}(\text{CN})_6]$
 (4) none is correct
65. Maximum magnetic moment is shown by :
 (1) d^5 (2) d^6
 (3) d^7 (4) d^8
66. Which of the following compounds is most sensitive of light ?
 (1) AgCl
 (2) AgCN
 (3) AgI
 (4) AgBr
67. The radius of La^{3+} is $1.06\text{\AA}$, which of the following given values will be closest to the radius of Lu^{3+}
 (At no. of $\text{Lu} = 71$, $\text{La} = 57$)
 (1) $1.6\text{\AA}$
 (2) $1.4\text{\AA}$
 (3) $1.06\text{\AA}$
 (4) $0.85\text{\AA}$

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| <p>68. Cerium ($Z = 58$) is an important member of the lanthanoids. Which of the following statements about cerium is incorrect -</p> <ol style="list-style-type: none"> (1) Cerium (IV) acts as an oxidising agent (2) The +3 oxidation state of cerium is more stable than the +4 oxidation state (3) The +4 oxidation state of cerium is not known in solutions (4) The common oxidation states of cerium are +3 and +4 <p>69. The lanthanoid contraction is responsible for the fact that -</p> <ol style="list-style-type: none"> (1) Zr and Y have about the same radius (2) Zr and Nb have similar oxidation state (3) Zr and Hf have about the same radius (4) Zr and Zn have similar oxidation state <p>70. Lanthanoid contraction is caused due to</p> <ol style="list-style-type: none"> (1) the same effective nuclear charge from Ce to Lu (2) the imperfect shielding on outer electrons by 4f electrons from the nuclear charge (3) the appreciable shielding on outer electrons by 4f electrons from the nuclear charge (4) the appreciable shielding on outer electrons by 5d electrons from the nuclear charge <p>71. Identify the incorrect statement among the following-</p> <ol style="list-style-type: none"> (1) d-block elements show irregular and erratic chemical properties among themselves (2) Gd has electronic configuration $4f^7 5d^1 6s^2$ (3) The chemistry of various lanthanoids is very similar (4) 4f and 5f-orbitals are equally shielded | <p>72. The actinoids exhibits more number of oxidation states in general than the lanthanoids. This is because -</p> <ol style="list-style-type: none"> (1) The 5f-orbitals are more buried than the 4f-orbitals (2) There is a similarity between 4f and 5f in the their angular part of the wave function (3) The actinoids are more reactive than the lanthanoids (4) The 5f-orbitals extend further from the nucleus than the 4f-orbitals <p>73. Larger number of oxidation states are exhibited by the actinides than those by the lanthanides, the main reason being</p> <ol style="list-style-type: none"> (1) 4f orbitals more diffused than the 5f orbitals (2) lesser energy difference between 5f and 6d than between 4f and 5d orbitals (3) more energy difference between 5f and 6d than between 4f and 5d orbitals (4) more reactive nature of the actinides than the lanthanides <p>74. Knowing that the chemistry of lanthanoids (Ln) is dominated by its +3 oxidation state, which of the following statements is incorrect ?</p> <ol style="list-style-type: none"> (1) Ln(III) compounds are generally colourless (2) Ln(III) hydroxides are mainly basic in character (3) Because of the large size of the Ln(III) ions the bonding in its compounds is predominantly ionic in character (4) The ionic sizes of Ln(III) decrease in general with increasing atomic number |
|---|---|

SPACE FOR ROUGH WORK

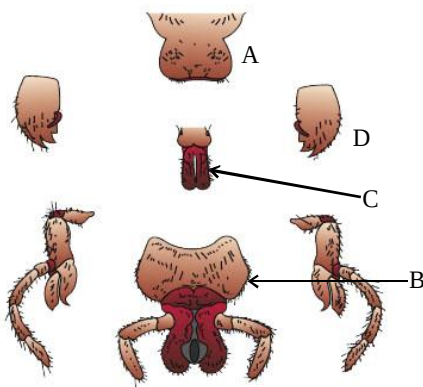
- | | |
|---|--|
| <p>75. In context of the lanthanoids, which of the following statements is not correct ?</p> <ol style="list-style-type: none"> (1) Because of similar properties the separation of lanthanoids is not easy (2) Availability of 4f electrons results in the formation of compounds in +4 state for all the members of the series (3) There is a gradual decrease in the radii of the members with increasing atomic number in the series (4) All the members exhibit +3 oxidation state <p>76. When rain is accompanied by a thunderstorm, the collected rain water will have a pH value</p> <ol style="list-style-type: none"> (1) slightly higher than that when the thunderstorm is not there (2) uninfluenced by occurrence of thunderstorm (3) which depends on the amount of dust in air (4) slightly lower than that of rain water without thunderstorm <p>77. The smog is essentially caused by the presence of :</p> <ol style="list-style-type: none"> (1) O_2 and O_3 (2) O_3 and N_2 (3) Oxides of sulphur and nitrogen (4) O_2 and N_2 <p>78. Regular use of which of the following fertilizers increases the acidity of soil ?</p> <ol style="list-style-type: none"> (1) potassium nitrate (2) Urea (3) Superphosphate of lime (4) Ammonium sulphate | <p>79. Identify the wrong statements in the following :-</p> <ol style="list-style-type: none"> (1) Chlorofluorocarbons are responsible for ozone layer depletion (2) Greenhouse effect is responsible for global warming (3) Ozone layer does not permit infrared radiation from the sun to reach the earth (4) Acid rains is mostly because of oxides of nitrogen and sulphur <p>80. What is DDT among the following :</p> <ol style="list-style-type: none"> (1) Greenhouse gas (2) A fertilizer (3) Biodegradable pollutant (4) Non-biodegradable pollutant <p>81. The gas leaked from a storage tank of the Union Carbide plant in Bhopal gas tragedy was:</p> <ol style="list-style-type: none"> (1) Methylisocyanate (2) Methylamine (3) Ammonia (4) Phosgene <p>82. Statement-1 : Nitrogen and Oxygen are the main components in the atmosphere but these do not react to form oxides of nitrogen.</p> <p>Statement-2 : Both CO_2 and CH_4 are green house gases.</p> <ol style="list-style-type: none"> (1) Both are correct statement (2) Only statement-1 is correct (3) Only statement-2 is correct (4) Both are incorrect statement |
|---|--|

SPACE FOR ROUGH WORK

83. The concentration of fluoride, lead, nitrate and iron in a water sample from an underground lake was found to be 1000 ppb, 40 ppb, 100 ppm and 0.2 ppm, respectively. This water is unsuitable for drinking due to high concentration of :
- (1) Fluoride (2) Lead
(3) Nitrate (4) Iron
84. A water sample has ppm level concentration of following anions $F^- = 10$; $SO_4^{2-} = 100$; $NO_3^- = 50$ the anion/anions that make / makes the water sample unsuitable for drinking is / are :
- (1) only NO_3^-
(2) both SO_4^{2-} and NO_3^-
(3) only F^-
(4) only SO_4^{2-}
85. Identify the pollutant gases largely responsible for the discoloured and lustreless nature of marble of the Taj Mahal
- (1) O_3 and CO_2
(2) CO_2 and NO_2
(3) SO_2 and O_3
(4) SO_2 and NO_2
86. Which of the following is a set of green house gases ?
- (1) O_3 , N_2 , CO_2 , NO_2
(2) CH_4 , O_3 , N_2 , SO_2
(3) CO_2 , CH_4 , N_2O , O_3
(4) O_3 , NO_2 , SO_2 , Cl_2
87. Biochemical Oxygen Demand (BOD) value can be a measure of water pollution caused by the organic matter. Which of the following statements is correct -
- (1) Anaerobic bacteria increase the BOD value
(2) Aerobic bacteria decrease the BOD value
(3) Polluted water has BOD value higher than 10 ppm
(4) Clean water has BOD value higher than 10 ppm
88. The recommended concentration of fluoride ion in drinking water is up to 1 ppm as fluoride ion is required to make teeth enamel harder by converting $[3Ca_3(PO_4)_2.Ca(OH)_2]$ to :
- (1) $[3(CaF_2).Ca(OH)_2]$
(2) $[3(Ca_3(PO_4)_2.CaF_2)]$
(3) $[3(Ca(OH)_2).CaF_2]$
(4) $[CaF_2]$
89. A water sample has ppm level concentration of the following metals :
 $Fe = 0.2$; $Mn = 5.0$; $Cu = 3.0$; $Zn = 5.0$.
The metal that makes the water sample unsuitable drinking is:
- (1) Zn (2) Fe (3) Mn (4) Cu
90. Which of the following conditions in drinking water causes methemoglobinemia ?
- (1) > 50 ppm of lead
(2) > 100 ppm of sulphate
(3) > 50 ppm of chloride
(4) > 50 ppm of nitrate

SPACE FOR ROUGH WORK

91. Head of cockroach is :-
 (1) Triangular and formed by the fusion of nine segments
 (2) Circular in shape and formed by the fusion of eight segments
 (3) Triangular in shape and formed by the fusion of six segments
 (4) Hexagonal in shape and formed by the fusion of nine segments
92. The abdomen of both males and females cockroach consist of :-
 (1) 9 segments (2) 10 segments
 (3) 7 segments (4) 8 segments
93. If we want to identify sex of cockroach which part of body should be observed ?
 (1) Head (2) Thorax
 (3) Abdomen (4) Neck
94. In *Periplaneta americana* exchange of gases take place at the :-
 (1) Trachea by diffusion
 (2) Tracheoles by diffusion
 (3) Main tracheal trunk
 (4) Atrium
95. Given below is a diagram of mouth parts of cockroach labelled A, B, C and D.



Find out the correct one :-

- (1) A-Labium, B-Labrum, C-Mandible
 D-Hypopharynx
 (2) A-Labium, B-Labrum, C-Hypopharynx
 D-Mandible
 (3) A-Labrum, B-Labium, C-Hypopharynx
 D-Mandible
 (4) A-Labrum, B-Labium, C-Mandible
 D-Hypopharynx

96. What is/are common in Earthworm and cockroach:-
 (1) Gizzard
 (2) Double, ventral and solid nerve cord
 (3) Enteronaphric nature of malpighian tubules of cockroach and septal nephridia of earthworm
 (4) All are correct
97. Brain of cockroach is represented by :-
 (1) Sub-oesophageal ganglia
 (2) Sub-pharyngeal ganglia
 (3) Supra-oesophageal ganglia
 (4) Supra-pharyngeal ganglia
98. Which one is incorrect about typhlosole in earthworm.
 (1) Typhlosole starts after 26th segment
 (2) Typhlosole is absent in last 23rd-25th segments of earthworm
 (3) It is internal median fold of dorsal wall of intestine
 (4) It is present in whole length of intestine i.e. 15th-120th segment
99. Egg of cockroach are :-
 (1) Centrolacithal and mesolacithal
 (2) Telolacithal and megalacithal
 (3) Centrolacithal and megalacithal
 (4) Telolacithal and microlacithal
100. When plenty of water is available then earthworms are :-
 (1) Ureotelic
 (2) Ammonotelic
 (3) Uricotelic
 (4) Excreat guanine
101. In *Pheretima posthuma* which of the following segment do not have setae :-
 (1) Second, 119th and clitellum
 (2) First, 2nd and last
 (3) First, last and clitellum
 (4) In pre clitellar segments (1st-13th)

102. Find out the correct statement for the blood vascular system of earthworm.

- (1) Pheretima exhibits an open type of blood vascular system, consisting of blood vessels, capillaries and heart
- (2) Haemoglobin is dissolved in blood plasma
- (3) Contraction in blood vessels keep blood circulation in both direction in all vessels
- (4) Blood is not confined to the heart and blood vessels

103. Cockroaches are brown or black bodied animals that are included in classA..... of phylumB..... .

Select correct words for A and B :

	A	B
(1)	Insecta	Arthropoda
(2)	Arthropoda	Insecta
(3)	Amphibia	Arthropoda
(4)	Both (1) and (3)	

104. Number of ommatidia in each compound eye in cockroach ?

- (1) 500
- (2) 1000
- (3) 2000
- (4) 4000

105. Which number of sternum is boat shaped in the female cockroach?

- (1) 9
- (2) 7
- (3) 6
- (4) 10

106. Match the following column of chitin plate in cockroach :-

Sclerite	Body side
(i) Sternum	(a) Lateral
(ii) Pleuron	(b) Ventral
(iii) Tergum	(c) Dorsal

- (1) i-a, ii-b, iii-c
- (2) i-b, ii-a, iii-c
- (3) i-c, ii-a, iii-b
- (4) i-b, ii-c, iii-a

107. Match the different structure in Periplaneta in their respective segment :-

Structures	Segments
(i) Anal cerci	(a) 2 nd -6 th segment
(ii) Anal style	(b) 6 th segment
(iii) Testes	(c) 10 th tergum in male and female
(iv) Ovary	(d) 9 th sternum in male
(v) Spermatheca	(e) 4 th , 5 th , 6 th segment
(vi) Mushroom gland	(f) 6 th -7 th segment

- (1) 1-d, ii-c, iii-e, iv-a, v-f, vi-b
- (2) i-c, ii-d, iii-f, iv-b, v-a, vi-e
- (3) i-c, ii-d, iii-e, iv-a, v-b, vi-f
- (4) i-c, ii-d, iii-a, iv-e, v-f, vi-b

108. Which blood vessel in earthworm distribute the blood to body wall, septa, septal nephridia and gonads :-

- (1) Dorsal vessel
- (2) Ventral vessel
- (3) Supra-oesophageal vessel
- (4) Lateral oesophageal vessel

109. Heart of cockroach is

- (1) 6 chambered
- (2) 9 chambered
- (3) 13 chambered
- (4) 26 chambered

110. Match the column A with column B for the *pheretima posthuma* :-

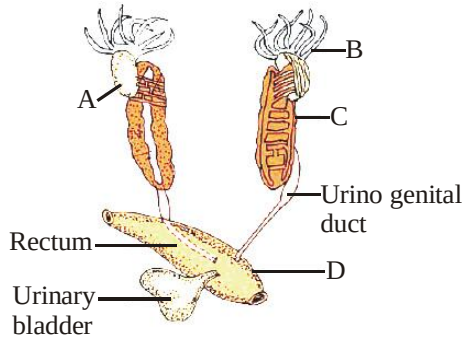
Column-A	Column-B
(i) Lateral hearts	(a) 26 th -22 nd segments
(ii) Blood glands	(b) 26 th -last segments
(iii) Lymph glands	(c) 7,9,12 and 13th segment
(iv) Intestinal caecae	(d) 4,5,6 segments

- (1) i-c, ii-d, iii-b, iv-a
- (2) i-d, ii-c, iii-a, iv-b
- (3) i-a, ii-b, iii-d, iv-c
- (4) i-c, ii-a, iii-b, iv-d

111. Alary muscles of coackroach are associated with :

- (1) Respiration
- (2) Circulation
- (3) Digestion
- (4) Locomotion

112. Following is the diagram of the male reproductive system of frog select the correct set of names labelled A,B,C and D :-



	A	B	C	D
(1)	Fat bodies	Ureter	Kidney	Cloaca
(2)	Testes	Ureter	Cloaca	Kidney
(3)	Adrenal gland	Bidder canal	Ureter	Testes
(4)	Testes	Fatbodies	Kidney	Cloaca

113. Ureter act as Urinogenital duct in :-
 (1) Frog's both male & female
 (2) Male frog
 (3) Female frogs
 (4) None of these
114. Frog are beneficial for man kind because they :
 (1) eat insects and protect the crop
 (2) maintain ecological balance
 (3) the muscular leg of frog are used as food
 (4) all of these
115. The frog's body is divisible into :-
 (1) Head,neck, abdomen
 (2) Head, neck, Trunk
 (3) Head, Trunk
 (4) Head, neck, Trunk & Tail
116. Which of the following statements is not true about frog ?
 (1) The body colour offers it protective colouration
 (2) Summer sleep of frog is called aestivation
 (3) Tail is present in Tadpole
 (4) Mouth is bounded by a pair of lips

117. Total number of vertebrae in frog is :-
 (1) 12 (2) 10
 (3) 26 (4) 33
118. The structure present in man but absent in frog is :
 (1) Pancrease (2) Salivary gland
 (3) Thyroid gland (4) Adrenal gland
119. Frog is :-
 (1) Homeothermic (2) Poikilothermic
 (3) Homeostatic (4) Worm blooded
120. Capacity of amphibians to change colour is called :-
 (1) Synchronous (2) Metachronous
 (3) Metachrosis (4) None of these
121. In male frog 10-12 vasa efferentia enter the kidney and open into :
 (1) Urinogenital duct
 (2) Seminal vesicle
 (3) Spermetophore
 (4) Bidder's canal
122. In female frog, the excretory system consists of :
 (1) Kidney and Ureters only
 (2) Kidney and Urinary bladder only
 (3) Kidney, Ureters, Urinary bladder and cloaca only
 (4) Kidney and cloaca only
123. The summer sleep of frog is termed as
 (1) Neoteny
 (2) Aestivation
 (3) Hibernation
 (4) Both 2 and 3 canal
124. Cerebral hemisphere is the part of (In frog):-
 (1) Fore brain
 (2) Mid brain
 (3) Hind brain
 (4) Spinal cord
125. Frog skull is :-
 (1) Noncondylic
 (2) Dicondylic
 (3) Monocondylic
 (4) None of these

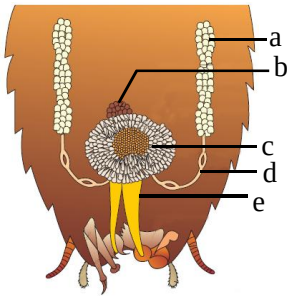
126. The middle ear of *Rana tigrina* has :-
 (1) 3 ear ossicles i.e. malleus, incus and stapes
 (2) One ear ossicle columella
 (3) 2 ear ossicle malleus & Incus
 (4) No ear ossicle
127. The forelimb of male frog is distinct from female frog due to the presence of :-
 (1) forearm (2) thumb
 (3) Web (4) Copulatory pad
128. Urinogenital duct of frog receives :-
 (1) Sperms (2) Urine
 (3) Faecal matter (4) Both (1) & (2)
129. A mature female frog can lay :-
 (1) 2500 to 3000 ova (2) 250 to 300 ova
 (3) 25 to 30 ova (4) 125 to 130 ova
130. Autonomic nervous system of frog consists of :-
 (1) Brain and spinal cord
 (2) Cranial and spinal nerves
 (3) Sympathetic nerve and para sympathetic nerve
 (4) Brain and cranial nerves
131. Spinal cord of frog extends from medulla oblongata through :-
 (1) Cerebellum
 (2) Foramen magnum
 (3) Cranial nerve
 (4) Diencephalon
132. Frog mainly excretes :-
 (1) Ammonia (2) Uric acid
 (3) Citrulline (4) Urea
133. In frogs urine is temporarily store in :-
 (1) Kidneys (2) Urinary bladder
 (3) Ureters (4) Cloaca
134. Truncus arteriosus/conus arteriosus is present on ____ side of heart of frog :-
 (1) Dorsal (2) Lateral
 (3) Ventral (4) Dorso-lateral
135. Which of the following acts as an respiratory organ in frog on land ?
 (1) Skin (2) Buccal cavity
 (3) Lungs (4) All of these
136. Air enters into the lungs of frog through the :-
 (1) Skin (2) Buccal cavity
 (3) Mouth (4) Cloaca
137. Which is the triangular structure associated with heart of frog present on dorsal side and receives the blood through major veins called vena cava ?
 (1) Conus arteriosus
 (2) Sinus venosus
 (3) Right atrium
 (4) Ventricle
138. What will happen if head of cockroach is cut off ?
 (1) No effect
 (2) It will die immediately
 (3) It will remain alive for few minutes
 (4) It will remain alive for a week
139. The head in cockroach shows great mobility in all directions due to :-
 (1) Flexible head capsule which also bears compound eyes
 (2) Flexible neck which is an extension of prothorax
 (3) Flexible neck which is an extension of mesothorax
 (4) Flexible neck which is an extension of metathorax
140. The mouth parts of cockroach are :-
 (1) Piercing and sucking type
 (2) Biting and cutting type
 (3) Chewing and biting type
 (4) Siphoning and sponging type
141. In cockroach, wings arise from :-
 (1) Prothorax and mesothorax and are two pairs
 (2) Mesothorax and metathorax and are two pairs
 (3) All the thoracic segments and are three pairs
 (4) First three abdominal segments and are three pairs

142. The abdomen & Thoracic segments in male cockroach are :-
 (1) 10 and 11 segments respectively
 (2) 3 and 10 segments respectively
 (3) 10 and 3 segments respectively
 (4) 10 and 9 segments respectively
143. Which of the following is incorrect for forewings of cockroach ?
 (1) Also called mesothoracic wings or tegmina
 (2) Opaque, dark and leathery
 (3) Cover the hind wings when at rest
 (4) Used in flight
144. In female cockroach, brood or genital pouch is formed by :-
 (1) 7th, 8th and 9th sternum
 (2) 7th sternum and 8th and 9th terga
 (3) 9th, 10th terga, and 9th sternum
 (4) 9th and 10th sternum, and 9th tergum
145. Read the following statements and find out the incorrect statement about cockroach :-
 (1) The blood (haemolymph) is composed of colourless plasma and haemocytes
 (2) Heart of cockroach consists of elongated muscular tube lying along mid ventral line of thorax and abdomen
 (3) The hindgut is broader than midgut and differentiated into ileum, colon and rectum
 (4) Both gizzard and mandible helps in grinding the food particles
146. Malpighian tubule are present at the junction of :-
 (1) Foregut and midgut, and helps in the secretion of digestive enzyme
 (2) Midgut and hindgut, and helps in the grinding the food particles
 (3) Foregut and midgut, and helps in the removal of excretory products from haemolymph
 (4) Midgut and hindgut, and helps in the removal of excretory products from haemolymph
147. The 10th segment bears a pair of jointed filamentous structure in cockroach which is called :-
 (1) Anal cerci, in both sexes
 (2) Anal style, in both sexes
 (3) Anal cerci, in male cockroach only
 (4) Anal style, in male cockroach only
148. With the help of several ommatidia, a cockroach can receive several images of an object. This kind of vision is called :-
 (1) Nocturnal vision, being common during night
 (2) Mosaic vision, with more sensitivity but less resolution
 (3) Mosaic vision, with more resolution but less sensitivity
 (4) Nocturnal vision, with more sensitivity and more resolution
149. On an average, female cockroach producesa..... oothecae, each containing ...b....eggs. The nymph grows by moulting aboutc..... times to reach the adult form.
 (1) a-9 to 10, b-14 to 16, c-13
 (2) a-14, to 16, b-9 to 10, c-13
 (3) a-9 to 10, b-15 to 40, c-10
 (4) a-2 to 20, b-14 to 16, c-13
150. Which is incorrect about cockroach ?
 (1) The anterior part of female genital pouch contains female gonopore, spermathecal pores and collateral glands
 (2) Male genital pouch contains ventral anus, dorsal male genital pore and gonapophysis
 (3) The heart is differentiated into funnel shaped chambers with ostia on either side
 (4) Blood vascular system is open type in which blood from sinuses enter heart through ostia and pumped anteriorly to sinuses again

151. Frog heart consist of :

- (1) Two auricles, one sinus venosus, one conus arteriosus and one ventricle
- (2) One auricle and two ventricles
- (3) Two auricles and two ventricles
- (4) One sinus venosus, one auricle & 2 ventricles

152. Recognise the figure and find out the correct matching.



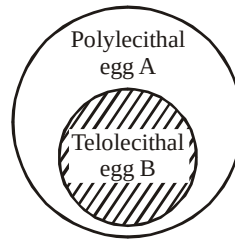
- (1) e-ejaculatory duct, d-vas deferens, a-testis, b-phallic gland
- (2) e-seminal vesicle, a-vas deferens, b-testis, d-phallic gland
- (3) a-seminal vesicle, b-vas deferens, e-testis, c-phallic gland
- (4) b-seminal vesicle, c-vas deferens, d-testis, e-phallic gland

153. Find out the correct match from the following table.

	Column-I	Column-II	Column-III
(i)	Alecithal	Yolk absent	Human
(ii)	Microlecithal	Yolk very small	Eutheria
(iii)	Mesolecithal	Moderate yolk	Amphibia

- (1) (i) & (ii)
- (2) (ii) & (iii)
- (3) (i) & (iii)
- (4) (i), (ii), (iii)

154.



In above diagram frog egg and bird egg can be placed respectively in :-

- (1) Circle 'A' and 'A'
- (2) Circle A and B
- (3) Both 1 & 2
- (4) Circle B and A

155. Find out the correct match from the table :-

	Column-I	Column-II	Column-III
(i)	Tertiary	Oocyte	Corona radiata
(ii)	Secondary	Ovary	Chorion
(iii)	Primary	Oviduct	Zona radiata

- (1) (ii) only
- (2) (i) & (ii)
- (3) (ii) & (iii)
- (4) (i) & (iii)

156. Which match is not correct ?

- (1) Cleidoic egg - birds
- (2) Centrolecithal egg - Reptile
- (3) Telolecithal - Amphibia
- (4) Discoidal - Prototherian

157. Which of the following statements are true about cleavage ?

- (a) K.I. increases during division
- (b) Both blastomeres and embryo size are reduced
- (c) Interphase stage only shows, s-phase
- (d) Yolk is inversely proportional to division

- (1) a, b, c, d
- (2) a, c, d
- (3) b, c, d
- (4) Only c and d

158. Which type of cleavage is present in alecithal egg of human ?

- (1) Determinate and equal holoblastic
- (2) Indeterminate and unequal holoblastic
- (3) Determinate and unequal meroblastic
- (4) Indeterminate and equal meroblastic

159. Which statement is wrong about human embryology ?
- (1) Zona pellucida disappears from blastocyst
 - (2) Implantation occurs in Gastrula stage
 - (3) Interstitial type of implantation
 - (4) 8 weeks embryo is called foetus
160. What is the correct sequence is followed in developing embryo ?
- (1) Ectoderm → Endoderm → Mesoderm
 - (2) Endoderm → Ectoderm → Extraembryonic mesoderm → Intraembryonic mesoderm
 - (3) Ectoderm → Intraembryonic mesoderm → Endoderm
 - (4) Chorion → Amnion → Endoderm → Ectoderm → Mesoderm
161. Gastrulation is characterised by :-
- (1) Morphogenetic movement of blastomeres
 - (2) Formation of three germinal layer
 - (3) (1) and (2) both
 - (4) Neural tube formation
162. Cleavage starts when zygote moves towards :
- (1) Ampulla to Isthmus
 - (2) Isthmus to Ampulla
 - (3) Infundibulum to Ampulla
 - (4) Ampulla to Infundibulum
163. If we could monitor the amount of total gonadotropin activity in pregnant women, we would expect :-
- (1) High levels of FSH and LH in the uterus to stimulate endometrial thickening
 - (2) High levels of circulating FSH and LH to stimulate implantation of the embryo
 - (3) High levels of HCG in the uterus to stimulate endometrial thickening
 - (4) High levels of circulating HCG to stimulate estrogen and progesterone synthesis
164. During cleavage the cell division is very rapid. The daughter cells do not undergo any growth and the cell thus become gradually smaller in volume. Hence :-
- (1) There is no growth in the volume and size of the embryo
 - (2) The embryo grows in volume and same in size
 - (3) The embryo becomes smaller in volume
 - (4) The embryo remains static
165. Read the following statement
- (a) After implantation, finger like projections appear on the trophoblast called chorionic villi
 - (b) The blastomeres in the blastocyst are arranged into an outer layer called trophoblast.
 - (c) Placenta also acts as an endocrine tissue.
 - (d) hCG, hPL and Relaxin are produced in women during pregnancy also.
- How many of the above statements are wrong ?
- (1) One
 - (2) Three
 - (3) Two
 - (4) Four
166. Arrange these reproductive events occurring in human from last to first and find out the correct sequence :-
 Insemination (A), Parturition (B), Formation of blastocyst (C), Gametogenesis (D), Gestation (E), Fertilisation (F)
- (1) D,A,F,C,E,B
 - (2) D,F,A,B,C,E
 - (3) B,E,C,F,A,D
 - (4) B,C,E,F,A,D
167. During the development of embryo which of the following occurs first-
- (1) Differentiation of tissue
 - (2) Differentiation of cell
 - (3) Differentiation of organs
 - (4) Differentiation of organ systems
168. How many of the following events occur in fallopian tube?
- | | |
|------------------|------------------|
| A. Fertilisation | B. Cleavage |
| C. Morulation | D. Organogenesis |
- (1) One
 - (2) Two
 - (3) Three
 - (4) Four

169. After implantation of blastocyst, some finger like projections appear on the surface of trophoblast called :-

- (1) Yolk sac villi
- (2) Chorionic villi
- (3) Cilia
- (4) Stereocilia

170. These are some events associated with reproduction and embryonic development. Arrange these from first to last and select the correct answer.

- (a) Start formation of corpus luteum
- (b) Cleavage
- (c) Fertilisation
- (d) Ovulation
- (e) Implantation
- (f) Formation of Graafian follicle
- (g) Blastocyst
- (h) Morula
- (i) Zygote

- (1) f, d, a, c
- (2) f, d, i, c
- (3) a, f, d, c
- (4) i, h, g, f

171. The first sign of growing foetus may be noticed by :-

- (1) Movement of foetus
- (2) Appearance of hair on head
- (3) Listening to the heart sound carefully through the stethoscope
- (4) Formation of limbs

172. In which stage of development the embryonic cells form the germinal layers by the movement?

- (1) Morula
- (2) Blastula
- (3) Gastrula
- (4) Zygote

173. Match the following and choose the correct answer:

(a)	Inner cell mass	(i)	Embedding of blastocyst in the endometrium
(b)	Cleavage	(ii)	Group of cells that would differentiate as embryo
(c)	Trophoblast	(iii)	Outer layer of blastocyst attached to the endometrium
(d)	Implantation	(iv)	Mitotic division of zygote

- (1) a-(ii), b-(i), c-(iii), d-(iv)
- (2) a-(iii), b-(iv), c-(ii), d-(i)
- (3) a-(ii), b-(iv), c-(iii), d-(i)
- (4) a-(iii), b-(i), c-(ii), d-(iv)

174. Match the column-A and B about the embryonic development of human :-

Column-A		Column-B	
(i)	End of one month	(a)	Most of the major organ systems developed
(ii)	End of second month	(b)	Appearance of hair on the head
(iii)	End of three month	(c)	Heart formation
(iv)	During fifth month	(d)	Eye-lids separated
(v)	End of sixth month	(e)	Limbs and digits formation

- (1) (i) – e, (ii) – b, (iii) – c, (iv) – a, (v) – d
- (2) (i) – e, (ii) – c, (iii) – b, (iv) – b, (v) – d
- (3) (i) – c, (ii) – e, (iii) – b, (iv) – a, (v) – d
- (4) (i) – c, (ii) – e, (iii) – a, (iv) – b, (v) – d

175. Which of the following is injected by the doctors to induce delivery in pregnant woman?

- (1) Estrogen
- (2) Cortisol
- (3) Relaxin
- (4) Oxytocin

- | | |
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| <p>176. Fertilization most commonly occurs if the ovum and sperms are transported simultaneously to</p> <ol style="list-style-type: none"> (1) Fimbriae-Infundibulum junction (2) Infundibulum-Ampulla junction (3) Ampulla (4) Isthmus-Fundus junction <p>177. The signals for parturition originate from the :-</p> <ol style="list-style-type: none"> (A) Fully developed foetus (B) Placenta (C) Uterus (D) Ovary <ol style="list-style-type: none"> (1) A & B (2) A & D (3) C & D (4) B & D <p>178. Which type of cells have the potency to give rise to all the tissue and organs in embryo ?</p> <ol style="list-style-type: none"> (1) Trophoblast (2) Amniogenic cells (3) Inner cell mass (4) Extra embryonic mesoderm | <p>179. How many of the following statements with regard to embryonic development in human are correct -</p> <ol style="list-style-type: none"> (I) In morula amount of cytoplasm and DNA is equal to Zygote (II) KI of blastomeres increases during cleavage (III) There is no division of cytoplasm during cleavage (IV) A three cell stage is found during cleavage <ol style="list-style-type: none"> (1) 4 (2) 1 (3) 3 (4) 2 <p>180. Which hormone is known as somatomammotropic hormone ?</p> <ol style="list-style-type: none"> (1) hCG (2) hPL (3) Oxytocin (4) Prolactin |
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Space For Rough Work
